



BEUTH HOCHSCHULE FÜR TECHNIK BERLIN
University of Applied Sciences

Master Colloquium

–

Project Guidelines

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2020

Example Projects from recent years

- Comparison of Motion Estimation Techniques in Video Coding Standards
- Face Detection / Number Plate Recognition Using OpenCV / MATLAB
- Feasibility Study of a Collision Detection by Means of Image Processing
- Remote Camera System Design
- Detecting and Decoding ADS-B Signals Using Software Defined Radio
- FM Transmission Using a Raspberry PI
- Technical objectives, requirements and solutions for 5G
- Comparison of Encryption Algorithms
- IoT architectures, protocols and opensource platforms based on 5G
- Overview of IoT Protocols, Smart City and Industrial Applications
- Overview of VPN, VPN's Protocol and Remote Acces VPN
- EEG Signal analysis for left/right hand classification
- kNN Classifier for Data Analytics
- Utilizing Machine Learning in Stock Market Prediction
- Low-cost free-space optical communication link

Project effort

To give you a rough idea of the effort to be put in the project

- 5 Credits = 150 working hours (h) per module
- 1 SWS attendance $\approx$ 20 h
- Presentation $\approx$ 10 h
- Report $\approx$ 10 h
- Roughly 100 h (2-3 weeks fulltime) left for working on the project

Project presentations and reports

Guidance for structuring presentation / report:

1. Context / Introduction / Ressources
 2. Target / Problem / Question
 3. Requirements / Criteria / Definitions
 4. Functional Designs / Concepts
 5. Implementation / Simulation / Analysis
 6. Tests / Verification / Results
 7. Outlook
- Depending on subject and timing, presentation may cover only bullets 1-4 / 7 and report would address all bullets.
 - For 1-year projects, the remaining bullets (5/6) can be addressed in the second part of the project.

Context / Introduction / Resources

- What is this used for?
- Where is it used?
- Why is it used?
- Why is it an interesting question?
- What is the history of all this?
- What is the motivation for this project?
- What resources do you have used (no quotation of references, rather general type of resources – Web, Personal Contacts, Companies, ...)?

Target / Problem / Question

- What is the target of the project?
- What problem is to be solved?
- What questions shall be answered?

Requirements / Criteria / Definitions

- How should the final system look like?
- What functions should be fulfilled?
- What constraints are accepted?
- What are criteria for comparison or evaluation (define them)?
- What aspects do you look at?
- How do you measure them?

Functional Design / Concepts

- What technical / theoretical approaches exist?
- What algorithms are known?
- Which tools, devices will be applied?
- What type of measurements will be made?
- How will the problem be solved / the question be answered?

Implementation / Simulation / Analysis

- Main part
- This is the actual work to be carried out
- It may be
 - the implementation in the device
 - the simulation of an algorithm to find optimum parameters
 - the theoretical comparison
 - the theoretical analysis of the problem
 - the collecting and analysing relevant data to carry out the comparison

Test / Verification / Results

- Proof that implementation / analysis works
- Measurements
- Mathematical proof
- Optimized parameters
- Comparison result
- ...

Outlook

- How does the project continue (if it is a larger project)?
- How could the project continue?

Project Definition

- Project Title
- Project Group (Names / IDs (*Matrikelnummern*))
- Short project description (some sentences)
- Input
 - What do you need to know beforehand?
 - What are the basics for this project?
- Output
 - What should be the result (no details)
- Tools (SW / HW)
- Dates (Registration, Start Project, End Project)
- Comments

Form is provided via moodle, please fill out and return to me!